3-D Superspectral and Superspatial–Spectral Incomplete Quaternization for Low-Rank Feature Extraction in Noisy Hyperspectral Image Classification

Longxia Qian, Xianyue Wang, Hongrui Wang, Mei Hong, and Yongchui Zhang

***Abstract*—Hyperspectral images (HSIs) acquired in real-world scenarios are frequently contaminated by various noise types, significantly compromising their classification accuracy. While low-rank feature extraction techniques can reconstruct clean data from noisy observations, most approaches either ignore spatial–spectral correlations or inefficiently process redundant spectral channels, ultimately restricting their classification performance. To address these issues, we propose a 3-D superspectral incomplete low-rank quaternization (3DSS-ILRQ) for robust HSI classification under diverse noise conditions. The proposed framework operates through two key mechanisms. First, the 3-D superspectral (3DSS) representation partitions the spectral bands of HSIs into heterogeneous groups, where each group contains homogeneous spectral bands. Second, the incomplete low-rank quaternization (ILRQ) operates adaptively on each group to extract low-rank features while retaining discriminative spectral cross-channel or spatial–spectral cross-element correlations via quaternion algebra, which structurally maintains the intragroup homogeneity and intergroup heterogeneity of spectral features. An extension of 3DSS-ILRQ, 3-D superspatial–spectral ILRQ (3DSSS-ILRQ), operates on heterogeneous spatial–spectral groups by 3-D superspatial–spectral (3DSSS) representation. Experimental validation on three benchmark datasets demonstrates that the proposed models outperform state-of-the-art methods under specific noise scenarios.**

***Index Terms*—Feature extraction, hyperspectral image (HSI) classification, incomplete quaternization, superunit.**

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I. Introduction

HYPERSPECTRAL image (HSI) classification has become crucial for long-distance high-precision ground observation, enabling precise identification of materials through their unique spectral signatures [1], [2]. However, the practical utility of HSI classification is fundamentally constrained by noise contamination, which is mainly an inevitable byproduct of sensor limitations and environmental variability during acquisition [3], [4], [5]. While low-rank recovery (LRR) methods have demonstrated potential in addressing this challenge, we argue that current approaches have primarily focused on data reconstruction quality while neglecting the critical needs of classification tasks [6], [7].

A. Evolution and Limitations of LRR Techniques

LRR-based HSI classification can be categorized into three methodological paradigms. The first category comprises matrix-based approaches that vectorize HSIs into 2-D matrices, inevitably compromising their intrinsic 3-D structural integrity [8], [9]. It is also worth noting that noise-induced performance degradation occurs in both 2-D spatial and 1-D spectral domains in HSIs, rather than just concentrated in the spatial domain of standard 2-D remote sensing images [10], [11]. Therefore, the LRR of HSIs necessitates fully considering joint spatial and spectral dimension information. The representation loss motivated the development of the second category, i.e., tensor decomposition methods including CP decomposition [12], Tucker decomposition [13], and tensor robust principal component analysis (TRPCA) [14], [15], which preserve spatial–spectral correlations through multilinear algebraic formulations. Nevertheless, these global tensor-based methods cannot protect essential spectral or spatial contextual information due to slice processing along any dimension [16], [17], [18].

The third category emerged through localized processing innovations, each with inherent constraints. In particular, superpixel-based techniques [19], [20] maintained spatial coherence at the expense of spectral correlation preservation.

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Quaternion-based LRR addresses the limitations of slice and unfolding processing and achieves spectral continuity by consolidating multiple genuine parts into a single quaternion part, but it is intrinsically limited to quaternionic band processing [21], [22]. Despite their computational power, deep-learning solutions [23] remained fundamentally constrained by these physical limitations. Crucially, all existing methods share a common weakness. They optimize the objective of general data reconstruction rather than classification-specific feature preservation.

B. Fundamental Challenges in Noisy HSI Classification

When applied to HSI classification, existing methods face several fundamental limitations arising from the distinctive characteristics of hyperspectral imagery. Remote sensing data exhibit distinct noise characteristics along both spatial and spectral dimensions. Sensor-induced artifacts (including striping and dead pixels) are spatially localized yet spectrally consistent, typically stemming from detector nonuniformity or calibration errors [24]. On the contrary, atmospheric noise is spectrally dependent but spatially uniform, with powerful effects at shorter wavelengths [25]. Conventional denoising techniques often struggle with this duality: spatial-only approaches (e.g., destriping filters [26]) ignore cross-band noise correlations, while spectral methods oversmooth spatially varying details [16], [17].

While adjacent spectral bands often exhibit substantial similarity (i.e., homogeneity) due to physical continuity in material reflectance properties, distant bands commonly show significant spectral variation (i.e., heterogeneity) [27], [28], [29]. This spectral heterogeneity poses a fundamental dilemma. Traditional global low-rank feature extraction methods [11], [12], [15] typically enforce uniform rank constraints across all bands, inevitably either oversmoothing discriminative features in heterogeneous regions or under-regularizing noisy homogeneous regions. Beyond heterogeneity, HSIs suffer from spectral redundancy. In particular, highly correlated adjacent bands provide minimal additional discriminative information yet significantly increase the computational burden and noise susceptibility. Existing methods may ignore this redundancy by processing all bands equally or employing naive band selection before processing, discarding potentially useful information.

The quaternion-based LRR methods [21], [22] offer an elegant solution for preserving cross-channel correlations without slice processing, but they face a critical dimensional constraint. Quaternion algebra only operates on 4-D data, while HSIs contain hundreds of spectral channels. Therefore, current quaternion methods address this mismatch by aggressive dimensionality reduction, which is designed to respect the quaternion's structural constraints while making adequate use of the spectral information.

Deep-learning-based feature extraction has recently gained significant attention due to its robust computational capabilities in processing complex data [30]. However, despite their advanced performance, these deep-learning algorithms still fail to account for the heterogeneity of distinct spectral channels or spatial–spectral elements [19], [31], [32]. Although the two-stream 2-D CNN [33] leverages spectral similarity to identify the most informative features, it still ignores the differences in spatial distribution. Conversely, Sun et al. [34] utilized superpixel segmentation to enhance spatial homogeneity, effectively handling mixed noise types, but they ignored spectral influences. Furthermore, most existing methods optimize for data reconstruction fidelity by minimizing the difference between denoised and clean images [33], whereas classification tasks fundamentally require discriminative features that maximize interclass separability rather than pixelwise accuracy. This objective mismatch results in suboptimal classification performance, as reconstruction-oriented denoising may preserve noise patterns that corrupt discriminative information or suppress subtle spectral variations critical for class distinction.

Above all, current manners remain fundamentally limited by their inability to simultaneously address spatial–spectral correlations, noise heterogeneity, spectral redundancy, and classification-oriented feature preservation. To bridge these gaps, we propose a 3-D superspectral incomplete low-rank quaternization (3DSS-ILRQ) framework designed explicitly for discriminative feature extraction in noisy HSI classification under redundant spectral channels. The synergy between a 3-D superspectral (3DSS) representation and incomplete low-rank quaternization (ILRQ) is motivated by three critical requirements. First, the spectral heterogeneity (varying correlation patterns across bands) demands adaptive partitioning rather than global processing. Second, quaternion algebra naturally preserves cross-channel correlations but is limited to four dimensions, necessitating spectral grouping. Third, classification requires discriminative features rather than reconstruction fidelity, justifying incomplete quaternization within homogeneous groups.

In particular, the 3DSS representation partitions spectral bands into heterogeneous groups (i.e., spectral superunits) containing homogeneous bands via controlled graphical matrixing, enabling the subsequent ILRQ to operate adaptively on each superunit while naturally eliminating redundant channels within each superunit. For scenarios requiring enhanced spatial–spectral coherence, we extend 3DSS-ILRQ to 3-D superspatial–spectral ILRQ (3DSSS-ILRQ), which simultaneously partitions both spatial and spectral dimensions into triply heterogeneous superunits. While 3DSS addresses spectral heterogeneity through band grouping, 3DSSS further handles spatial heterogeneity by incorporating spatial-row and spatial-column partitioning, creating structure-protected subcubes that preserve 2-D spatial arrangements. The flowchart of the proposed 3DSS-ILRQ and 3DSSS-ILRQ is demonstrated in Fig. 1, where the green boundary lines in the middle panel delineate the superunits, partitioning the HSI into multiple subcubes that are subsequently processed by ILRQ.

The main contributions of this article are outlined below.

1) We develop a 3DSS representation that extends superunit recognition from the spatial to the spectral domain through single-dimension controlled graphical matrixing, enabling adaptive partitioning of heterogeneous spectral bands while preserving the 3-D structure. This overcomes the limitation of spatial-only superunit

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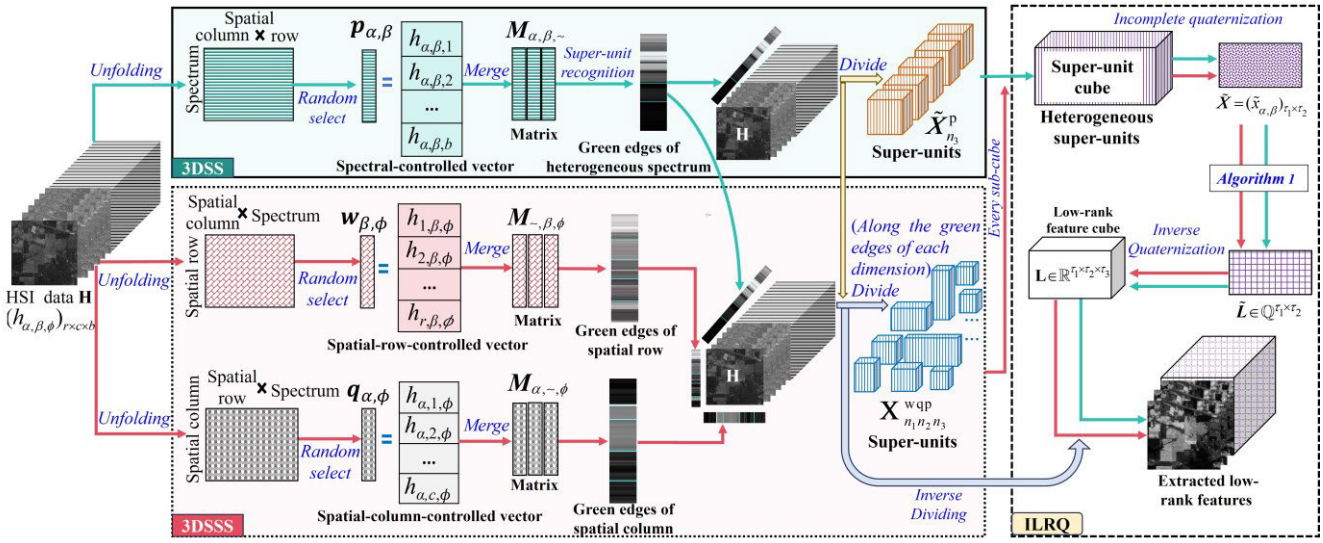


Fig. 1. Flowchart of the proposed 3DSS-ILRQ and 3DSSS-ILRQ frameworks. The upper panel shows 3DSS-ILRQ using a spectral-controlled vector for spectral superunit recognition. The lower panel shows 3DSSS-ILRQ using three controlled vectors for comprehensive spatial–spectral superunit recognition. Each superunit cube undergoes ILRQ in the right panel.

techniques that ignore spectral heterogeneity without accounting for spectral heterogeneity across different wavelength ranges.

2) We propose an ILRQ framework that synergistically integrates with 3DSS representation to reconcile the dimensional mismatch between quaternion algebra (four-dimensional) and high-dimensional HSI data. By adaptively operating on homogeneous spectral groups obtained from 3DSS, ILRQ naturally eliminates redundant channels while preserving discriminative cross-channel correlations, effectively leveraging abundant spectral information without violating the quaternion's structural constraints.
3) We extend 3DSS-ILRQ to 3DSSS-ILRQ by further incorporating spatial-row and spatial-column controlled graphical matrices, generating triply homogeneous superunits that simultaneously respect spatial and spectral heterogeneity. This advancement preserves the 2-D spatial structure while enabling spatial–spectral adaptive processing.

The remaining sections are demonstrated as follows. Section II reviews related work to highlight the technical advancements of our proposed models. Section III elaborates on the proposed methodology, comprehensively explaining its underlying principles and implementation process. Section IV presents and analyzes experimental results through a series of rigorous evaluations. Finally, Section V concludes this article and outlines potential directions for future research.

II. Related Work

A. Quaternization on Multichannel Images

Quaternization has demonstrated strong performance in multichannel images by preserving interchannel relationships critical for geological interpretation. This approach encodes red, green, and blue channels into quaternion imaginary components to maintain spectral coherence in traditional color image processing. For instance, Xu et al. [35] leveraged k-quaternion dictionary learning to enhance sparse representations in mineral mapping applications. Similarly, Gai and Huang [36] combined reduced biquaternions with deep networks to improve computational efficiency while preserving diagnostic spectral features for lithological classification. However, extending quaternization to high-dimensional HSI introduces fundamental challenges in preserving spectral fidelity. As the number of spectral channels increases, most quaternization methods require spectral band selection [37] or feature extraction [38] to compress hundreds of spectral channels into no more than four variables to adapt to the quaternion structure. As a result, it compromises the inherent advantage of abundant spectral features in multispectral [39] and hyperspectral data [40]. This limitation hinders the practical application of quaternization in tasks requiring detailed spectral information.

On the contrary to the abovementioned studies, our proposed model preserves crucial spectral features while eliminating redundant channels without additional compression. This enhancement is achieved through ILRQ, which is applied to subsets of homogeneous units rather than the entire dataset. Furthermore, it is important to distinguish our classification-oriented feature extraction framework from conventional noise reduction methods. While reconstruction-oriented denoising approaches [3], [16], [29] optimize for pixelwise fidelity by minimizing reconstruction error (e.g., PSNR, SSIM), they often oversmooth discriminative features critical for classification. On the contrary, our ILRQ framework targets discriminative feature extraction that maximizes interclass separability rather than reconstruction accuracy. Furthermore, HSI classification is affected by multiple factors beyond noise, including spectral variability (intraclass spectral variations caused by illumination, topography, and material heterogeneity [41]), spatial heterogeneity, and spectral

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redundancy. Our superunit-based adaptive processing inherently addresses these factors simultaneously by partitioning the HSI into heterogeneous groups and applying localized feature extraction.

B. Superunit Recognition on High-Dimensional Data

Superpixel segmentation is a common technique for superunit recognition in 2-D images, where irregular boundaries delineate distinct objects and separate targets from the background [19]. This approach reduces computational complexity by grouping pixels into superunits while enhancing representation capability. However, applying superpixel segmentation to high-dimensional data, such as HSIs, typically involves additional preprocessing steps, such as principal component analysis (PCA), and is limited to the spatial domain [27]. This restricts its ability to exploit the spectral information in HSIs fully. Despite its advantages, superunit recognition in HSIs faces two significant challenges. First, while it successfully identifies heterogeneous pixel groups for differentiated treatment, it often fails to preserve the original irregular 2-D structure. This limitation hinders the subsequent construction of tensor and quaternion data. Although Qian et al. [28] proposed block reconstruction to restore tensor structures, the plane spatial structure remains compromised. Second, existing superunit recognition methods overlook spectral homogeneity, which is critical for precise recognition in HSIs. These limitations motivate the development of our 3DSS and 3DSSS representations.

Our 3DSS representation extends superunit recognition from the spatial domain to the spectral domain, enabling the identification of homogeneous and heterogeneous spectral channels. This approach preserves the original 2-D spatial structure while capturing spectral homogeneity, addressing the limitations of traditional superpixel methods. Furthermore, the extended 3DSSS representation introduces spatial and spectral tripling superunits, which generate structure-protected homogeneous spatial–spectral subcubes. These innovations significantly enhance the robustness and accuracy of superunit recognition in high-dimensional data.

III. Proposed Methodology

A. Problem Description and Overview

As a high-dimensional remote sensing dataset, HSI can be presented as a 3-D cube $\boldsymbol{H} \in \mathbb{R}^{r\times c\times b}$ where r and c denote the spatial dimensions, and b represents the spectral dimension. In the HSI, the 2-D spatial pixel set can be presented as $\boldsymbol{P}_k \in \mathbb{R}^{r\times c}$ $(k = 1, 2, \ldots, b)$ containing a total of $s = r \times c$ objective pixels. To reduce indeterminate noise on real-world HSIs, the objective function of most general LRR models can be expressed as

$$\min_{\mathbf{L},\mathbf{N}} \left(f_a(\mathbf{L}) + \gamma f_b(\mathbf{N})\right), \quad \text{s.t. } \mathbf{H} = \mathbf{L} + \mathbf{N} \tag{1}$$

where $\mathbf{N} \in \mathbb{R}^{r\times c\times b}$ and $\mathbf{L} \in \mathbb{R}^{r\times c\times b}$ are the noisy component and the clean image of $\mathbf{H} \in \mathbb{R}^{r\times c\times b}$, respectively. $f_a(\cdot)$ and $f_b(\cdot)$ are two commonly different norm functions. γ is the parameter to regularize distinct norms.

As discussed in Sections I and II, we focus on uncovering the homogeneity and heterogeneity in the spectral and spatial domains of HSIs by perceptual identification. In addition, a high-dimensional-adapted quaternization LRR is supposed to preserve the essential information of the original colossal structure. However, when superunit identification is directly integrated with quaternization LRR, several challenges must be addressed: 1) how to conduct perceptual recognition within the spectral domain and maintain 3-D context and structure? 2) how to establish an appropriate quaternion LRR on high-dimensional spectral channels that exceed the natural capacity of quaternions; and 3) how to effectively operate on HSIs without considering redundant variables while retaining the inherent physical significance for further analysis?

To solve problem 1), we build a general framework based on the imaging mechanism of remote sensing HSIs, as shown in Fig. 1 and described as follows.

1) *Step 1:* For 3DSS representation, the spectral-dimension-controlled graphical matrix is built by repeating spectral-controlled vectors. Each vector retains all spectral channels of a randomly selected spatial pixel. For 3DSSS representation, spatial-row-controlled and spatial-column-controlled graphical matrices are constructed to preserve 2-D spatial structures.
2) *Step 2:* Inspired by entropy rate superpixel segmentation [19], 3DSS or 3DSSS representations operate on superunits to generate connected 3-D subcubes from the HSI data. This step divides the large data cube into smaller, manageable units, laying the foundation for noise removal.
3) *Step 3:* To solve problems 2) and 3), ILRQ is independently and adaptively applied to each subcube due to 3DSS and 3DSSS remodeling. This method considers the differences in various spatial–spectral superunits and removes noise while maintaining inherent 3-D structure and context. Moreover, it efficiently operates on a subset of homogeneous subcubes, naturally dropping some redundant spectra.

B. Single-Dimension Controlled Graphical Matrixing

As previously mentioned, there are typically strong correlations among adjacent spectral channels in HSIs, as illustrated in Fig. 2(a) and (b). It is evident that similar bands tend to cluster together, and many neighboring bands show high correlations. Therefore, it is possible to recognize spectral superunits perceptually. Identical to recognizing superunits of the 2-D spatial domain by emphasizing spatial pixel variability through spectral compression transformation, identifying connected superspectral units in HSIs involves conducting 2-D spectrum graphical reconstruction to highlight spectral band variability. The structured arrangement of pixels allows for directly incorporating spectral compression into the superunit identifier. However, unlike the spatial domain, the spectral dimension lacks prior 2-D structural information. Increasing the number of compressed or original pixel elements to create a 2-D spectral structure for superunit recognition affects the spectrum's variability, hindering smooth identification. To

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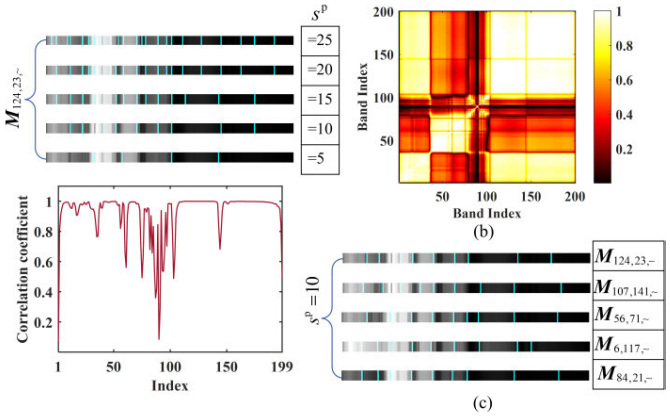


Fig. 2. (a) Superspectral-unit recognition of different s^p on the same $\mathbf{M}_{124,23,\sim}$ (upper) and correlation coefficient among adjacent bands in the IPs image (lower). (b) Heatmap of correlations among spectral bands. (c) Superspectral-unit recognition of $s^p = 10$ on different $\mathbf{M}_{\alpha,\beta,\sim}$.

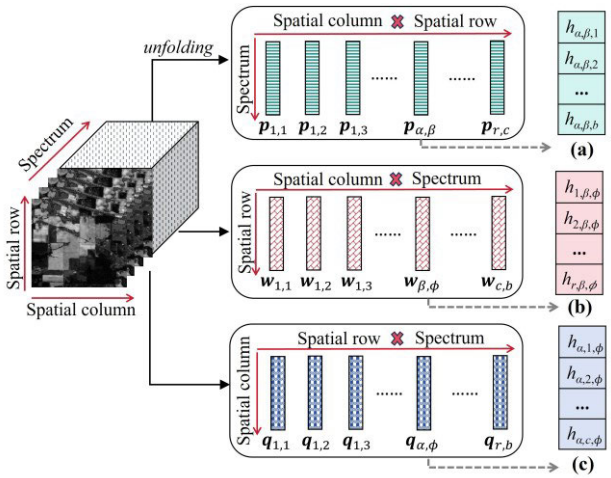


Fig. 3. Processes of obtaining single-dimension controlled vectors. (a) Spectra-controlled, (b) spatial-row-controlled, and (c) spatial-column-controlled vectors.

address these issues, we design a single-dimension controlled graphical matrix composition method for subsequent representations, as detailed below.

For 3DSS representation, the main objective of the spectral-dimension-controlled graphical matrix is to generate a 2-D graphical matrix that accentuates spectral variability while minimizing the number of pixels. Essentially, the spectral-dimension-controlled graphical matrix consists of repetitive spectral-controlled vectors built by maintaining global spectral channels and a randomly selected spatial pixel.

In particulat, the pixel vector in the αth spatial row and the βth spatial column of $\mathbf{H} = (h_{\alpha,\beta,\phi})_{r\times c\times b} (\alpha = 1,\ldots,r; \beta = 1,\ldots,c; \phi = 1,\ldots,b)$ can be represented as $\boldsymbol{p}_{\alpha,\beta} \in \mathbb{R}^b$, which is the (α,β)th spectral-dimension controlled vector that holds all spectral channels. Similarly, the (β,ϕ)th spatial-row-controlled vector and the (α,ϕ)th spatial-column-controlled vector can be defined as $\boldsymbol{w}_{\beta,\phi} \in \mathbb{R}^r$ and $\boldsymbol{q}_{\alpha,\phi} \in \mathbb{R}^c$, respectively. The generating processes of distinct controlled vectors are illustrated in Fig. 3.

To achieve superunit recognition in the spectral domain, our composition method maintains the global spectral channels as the column elements of the graphical matrix. At the same time, a randomly selected spatial pixel is repeated k times as the row elements. This variable control strategy emphasizes the characteristics of the target variable itself by managing the differences between the remaining variables to mitigate the negative impact of pixel variations on spectral homogeneity and heterogeneity discrimination during the recognition process. Simultaneously, the 2-D structural requirements of the superunit identifier for the input graphical matrix are also satisfied. The spectral-controlled graphical matrix $\boldsymbol{M}_{\alpha,\beta,\sim}$ constructed on $\boldsymbol{p}_{\alpha,\beta}$ is denoted as follows:

$$\boldsymbol{M}_{\alpha,\beta,\sim} = [\underbrace{\boldsymbol{p}_{\alpha,\beta}, \boldsymbol{p}_{\alpha,\beta}, \ldots, \boldsymbol{p}_{\alpha,\beta}}_{k}]^T \in \mathbb{R}^{k\times b} \tag{2}$$

where the superscript ^{T} represents the transpose operation. It should be noted that the value of k can be adjusted within a specific range. In principle, the central role of a superunit identifier is to delineate the boundary curve between different regions in the image. Creating the graphical matrix that governs pixel variability ensures that the boundaries of the bands are distinct and linear. Replicating the chosen spectral-controlled vector multiple times will not significantly alter the recognition results. However, an extensive input matrix will add to the computational load. Therefore, selecting k from $\{2, 3, 4, 5, 6\}$ is advisable.

Unlike 3DSS, the 3DSSS representation is supposed to operate on spectral and spatial superunits while maintaining the 3-D context and structure of HSIs. To break through the limitation of current superpixel segmentation that destroys the 2-D spatial structure, the spatial-row-controlled graphical matrix $\boldsymbol{M}_{\sim,\beta,\phi}$ constructed on $\boldsymbol{w}_{\beta,\phi}$ and the spatial-column-controlled graphical matrix $\boldsymbol{M}_{\alpha,\sim,\phi}$ constructed on $\boldsymbol{q}_{\alpha,\phi}$ are developed as

$$\boldsymbol{M}_{\sim,\beta,\phi} = [\underbrace{\boldsymbol{w}_{\beta,\phi}, \boldsymbol{w}_{\beta,\phi}, \ldots, \boldsymbol{w}_{\beta,\phi}}_{k}]^T \in \mathbb{R}^{k\times r} \tag{3}$$

$$\boldsymbol{M}_{\alpha,\sim,\phi} = [\underbrace{\boldsymbol{q}_{\alpha,\phi}, \boldsymbol{q}_{\alpha,\phi}, \ldots, \boldsymbol{q}_{\alpha,\phi}}_{k}]^T \in \mathbb{R}^{k\times c}. \tag{4}$$

C. 3DSS Representation and 3DSSS Representation

After creating a composite single-dimension controlled graphical matrix, the representation of 3DSS can effectively distinguish between uniform and varied spectral channels perceptually. This process generates interconnected smaller cubes of different scales from the original HSIs while preserving the structure. Initially, the spectral-controlled graphical matrix $\boldsymbol{M}_{\alpha,\beta,\sim}$ is divided into several interconnected regions with the edges of heterogeneous regions marked in green (refer to Fig. 1). Given the corresponding vertex set $\boldsymbol{\Upsilon}_{\alpha,\beta,\sim}$, the edge set $\boldsymbol{E}_{\alpha,\beta,\sim}$, and the edge subset $\boldsymbol{A}_{\alpha,\beta,\sim} \subseteq \boldsymbol{E}_{\alpha,\beta,\sim}$, the objective function of the superunit recognition on $\boldsymbol{M}_{\alpha,\beta,\sim}$ with entropy rate term function $R(\cdot)$ and balancing term function $B(\cdot)$ can be represented as

$$\underset{\boldsymbol{A}_{\alpha,\beta,\sim}\subseteq\boldsymbol{E}_{\alpha,\beta,\sim}}{\operatorname{argmax}} \left(tr\left(R\left(\boldsymbol{A}_{\alpha,\beta,\sim}\right)\right) + \chi \boldsymbol{B}\left(\boldsymbol{A}_{\alpha,\beta,\sim}\right)\right) \tag{5}$$

where $tr(\cdot)$ is the trace function. The parameter χ is adopted to make a balance of different terms, respectively. This

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function is derived from the superspatial-unit recognition in [19], which can adjust superspectral units to have similar sizes. By employing the single-dimension controlled graphical matrix composition approach, the output edges among various regions are aligned along the spectral dimension. Spectral channels within the same homogeneous superunit are exclusively dedicated to a single cluster based on the recognition results. This fulfills the requirements for subsequent analysis and significantly reduces the operational workload. To be specific, assume that $\boldsymbol{M}_{\alpha,\beta,\sim}$ can be divided into s^{p} superunits, where the n_3th superunit $\boldsymbol{S}^{\mathrm{p}}_{n_3}$ contains $\kappa^{\mathrm{p}}_{n_3}$ spectral channels as

$$\boldsymbol{M}_{\alpha,\beta,\sim} = \bigcup_{n_3=1}^{s^{\mathrm{p}}} \boldsymbol{M}_{\alpha,\beta,\sim}\left(\sum_{\psi=1}^{n_3-1}\kappa^{\mathrm{p}}_{\psi}+1:\sum_{\psi=1}^{n_3}\kappa^{\mathrm{p}}_{\psi}\right) = \bigcup_{n_3=1}^{s^{\mathrm{p}}} \boldsymbol{S}^{\mathrm{p}}_{n_3}$$
$$\text{s.t.} \sum_{n_3=1}^{s^{\mathrm{p}}} \kappa^{\mathrm{p}}_{n_3} = b. \tag{6}$$

According to the composition mechanism of $\boldsymbol{M}_{\alpha,\beta,\sim}$, the spectral channels can be divided into s^{p} subsets, where each subgroup $\boldsymbol{\Gamma}_n$ composed of $\kappa^{\mathrm{p}}_{n_3}$ spectral channels. Assume that each spectral variable can be represented as $\boldsymbol{\xi}_{\phi}$ with the index $\phi(\phi = 1,\ldots,b)$, the whole spectral set $\{\boldsymbol{\xi}_{\phi}|\phi = 1,\ldots,b\}$ can be defined as $\{\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3}|n_3 = 1,\ldots,s^{\mathrm{p}}\}$, where $\boldsymbol{\Gamma}_{n_3}$ is defined as

$$\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3} = \left[\boldsymbol{\xi}_{\sum_{\psi=1}^{n_3-1}\kappa^{\mathrm{p}}_{\psi}+1}, \boldsymbol{\xi}_{\sum_{\psi=1}^{n_3-1}\kappa^{\mathrm{p}}_{\psi}+2}, \ldots, \boldsymbol{\xi}_{\sum_{\psi=1}^{n_3}\kappa^{\mathrm{p}}_{\psi}}\right]. \tag{7}$$

The subsequent remodeling process based on the recognition results can protect the 3-D structure with spectral heterogeneity. As shown in Fig. 2(a), the boundary of the diverse spectrums, which marks the spectrum collection's separation, is near the mutation point on the correlation coefficient curve among adjacent bands. Furthermore, while the separation points obtained through the superspectral-unit recognition on the matrix $\boldsymbol{M}_{\alpha,\beta,\sim}$ established using random pixel vectors $\boldsymbol{p}_{\alpha,\beta}$ may differ, they typically fluctuate around the mutation point without significant variances. This demonstrates the reliability and resilience of the proposed segmentation algorithm. Consequently, the objective structure-protected homogeneous subcubes $\mathbf{X}^{\mathrm{p}}_{n_3}(n_3 = 1,\ldots,s^{\mathrm{p}})$ with varying scales for 3DSS representation (refer to Fig. 1) are

$$\mathbf{H} = \bigcup_{n_3=1}^{s^{\mathrm{p}}} \mathbf{H}\left(:,:,\sum_{\psi_3=1}^{n_3-1}\kappa^{\mathrm{p}}_{\psi_3}+1:\sum_{\psi_3=1}^{n_3}\kappa^{\mathrm{p}}_{\psi_3}\right) = \bigcup_{n_3=1}^{s^{\mathrm{p}}} \mathrm{X}^{\mathrm{p}}_{n_3}. \tag{8}$$

In the same way, 3DSSS representation needs to operate independently and adaptively on graphical matrices $\boldsymbol{M}_{\alpha,\beta,\sim}$, $\boldsymbol{M}_{\sim,\beta,\phi}$, and $\boldsymbol{M}_{\alpha,\sim,\phi}$. By repeatedly following the steps from (5) to (7), we can create structure-protected interconnected subcubes $\mathbf{X}^{\mathrm{wqp}}_{n_1n_2n_3}(n_1 = 1,\ldots,s^{\mathrm{w}}; n_2 = 1,\ldots,s^{\mathrm{q}}; n_3 = 1,\ldots,s^{\mathrm{p}})$ that rely on the spatial and spectral tripling superunits by:

$$\begin{aligned}\mathbf{H} &= \bigcup_{n_3=1}^{s^{\mathrm{p}}}\bigcup_{n_1=1}^{s^{\mathrm{w}}}\bigcup_{n_2=1}^{s^{\mathrm{q}}} \mathrm{X}^{\mathrm{p}}_{n_3} \\ &\quad\left(\sum_{\psi_1=1}^{n_1-1}\kappa^{\mathrm{w}}_{\psi_1}+1:\sum_{\psi_1=1}^{n_1}\kappa^{\mathrm{w}}_{\psi_1}, \sum_{\psi_2=1}^{n_2-1}\kappa^{\mathrm{q}}_{\psi_2}+1:\sum_{\psi_2=1}^{n_2}\kappa^{\mathrm{q}}_{\psi_2}\right) \\ &= \bigcup_{n_1=1}^{s^{\mathrm{w}}}\bigcup_{n_2=1}^{s^{\mathrm{q}}}\bigcup_{n_3=1}^{s^{\mathrm{p}}} \mathrm{X}^{\mathrm{wqp}}_{n_1n_2n_3}\end{aligned} \tag{9}$$

where $\kappa^{\mathrm{w}}_{n_1}$ and $\kappa^{\mathrm{q}}_{n_2}$ are the numbers of elements in the n_1th spatial superrow-unit and the n_2th spatial supercolumn-unit of $\boldsymbol{M}_{\sim,\beta,\phi}$ and $\boldsymbol{M}_{\alpha,\sim,\phi}$, respectively. So far, the representations of 3DSS and 3DSSS have made significant strides in overcoming the limitations of current superunit recognition. This breakthrough preserves the 2-D spatial structure and extends the identification of perceptual superunits beyond the spatial domain, laying the groundwork for ILRQ.

D. ILRQ on Homogeneous Subcubes

Quaternions have gained significant attention in image LRR due to their ability to preserve 2-D spatial structure and cross-spectral-channel correlations. To maintain the spatial context and correlation of homogeneous spectral variables in HSIs, we apply quaternization to the obtained subcubes, adapting to and addressing the heterogeneity of distinct subcubes to reduce noise. The first step in this process involves converting the 3-D subcubes into corresponding quaternions. Despite appearing to alter the 3-D structure, this process retains the context information of the homogeneous spectrum through reversible quaternization processing.

The previous superunit construction does not guarantee that the number of spectral channels in each subblock is less than four. Considering that not all spectra will be used in subsequent research, the high dimensionality of HSIs, and the similarity and redundancy of spectral variables, we propose an incomplete quaternization transformation. For instance, on the subcube $\mathrm{X}^{\mathrm{p}}_{n_3}$, the matching quaternionic matrix $\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3} \in \mathbb{Q}^{r\times c}$ is presented as

$$\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3} = \begin{bmatrix} \tilde{x}^{n_3}_{1,1} & \tilde{x}^{n_3}_{1,2} & \cdots & \tilde{x}^{n_3}_{1,c} \\ \tilde{x}^{n_3}_{2,1} & \tilde{x}^{n_3}_{2,2} & \cdots & \tilde{x}^{n_3}_{2,c} \\ \cdots & \cdots & \cdots & \cdots \\ \tilde{x}^{n_3}_{r,1} & \tilde{x}^{n_3}_{r,2} & \cdots & \tilde{x}^{n_3}_{r,c} \end{bmatrix} = \left(\tilde{x}^{n_3}_{\alpha,\beta}\right)_{r\times c} \tag{10}$$

s.t. $\tilde{x}^{n_3}_{\alpha,\beta}$, as shown at the bottom of the next page, where $\theta^{\mathrm{p}}_{n_3} = \sum_{\psi=1}^{n_3}\kappa^{\mathrm{p}}_{\psi}$ is the last spectral index of $\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3}$, $(\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3})^{t/100}$ returns the supremum of integers of t percentiles of the spectral index values of $\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3}$, and $\{u,v,\delta\}$ represents the set of basis complex operators. The incomplete operation is carried out on a subset of each homogeneous group, retaining essential features with a wide distribution range of wavelengths and eliminating redundant ones. Then, ILRQ is independently performed on each element in $\{\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}|n_3 = 1,\ldots,s^{\mathrm{p}}\}$. On the n_3th quaternion $\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}$, the objective function of ILRQ is given as follows:

$$\min_{\tilde{\boldsymbol{L}}^{\mathrm{p}}_{n_3},\tilde{\boldsymbol{N}}^{\mathrm{p}}_{n_3}} \left(\left\|\tilde{\boldsymbol{L}}^{\mathrm{p}}_{n_3}\right\|_{\dagger} + \lambda^{\mathrm{p}}_{n_3}\left\|\tilde{\boldsymbol{N}}^{\mathrm{p}}_{n_3}\right\|_{2,1}\right), \quad \text{s.t. } \tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3} = \tilde{\boldsymbol{L}}^{\mathrm{p}}_{n_3} + \tilde{\boldsymbol{N}}^{\mathrm{p}}_{n_3} \tag{11}$$

where $\lambda^{\mathrm{p}}_{n_3} = (\sqrt{\max(r,c)}\kappa^{\mathrm{p}}_{n_3})^{-1}$ is the regularization parameter; $\tilde{\boldsymbol{L}}^{\mathrm{p}}_{n_3}$ and $\tilde{\boldsymbol{N}}^{\mathrm{p}}_{n_3}$ refer to the required low-rank quaternion and the removed noisy part of the original $\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}$, respectively. $\|\cdot\|_{2,1}$ is the $l_{2,1}$-norm of the input matrix in a quaternionic complex space [35], [36]. In particular, the $l_{2,1}$-norm on $\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}$ equals $\sum_{\alpha=1}^{r}\sqrt{\sum_{\beta=1}^{c}|\tilde{x}^{n_3}_{\alpha,\beta}|^2}$ where $|\cdot|$ represents a modulus operator. Besides, the trace-norm on $\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}$ can be expressed as $\|\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}\|_{\dagger} = \sum\sigma(\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3})$, where $\sigma(\cdot)$ is the singular value of the input quaternion matrix. Meanwhile, the Frobenius norm is

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$\|\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}\|_{\mathrm{F}} = \sqrt{\sum_{(\sigma(\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}))^2}}$. It should be noted that the incomplete quaternization on the homogeneous subcubes $\mathrm{X}^{\mathrm{wqp}}_{n_1n_2n_3}$ from the 3DSSS representation is similar to the transformation on $\mathrm{X}^{\mathrm{p}}_{n_3}$ generated by the 3DSS representation. No matter which manner is produced, any input subcube can be rewritten as $\mathrm{X} = (h_{\alpha,\beta,\phi})_{\tau_1\times\tau_2\times\tau_3}$ and transformed to $\tilde{\boldsymbol{X}} = (\tilde{x}_{\alpha,\beta})_{\tau_1\times\tau_2}$. In this situation, (11) can be correspondingly generalized to

$$\min_{\tilde{\boldsymbol{L}},\tilde{\boldsymbol{N}}} \left(\left\|\tilde{\boldsymbol{L}}\right\|_{\dagger} + \lambda \left\|\tilde{\boldsymbol{N}}\right\|_{2,1} \right), \quad \text{s.t.} \tilde{\boldsymbol{X}} = \tilde{\boldsymbol{L}} + \tilde{\boldsymbol{N}} \tag{12}$$

where $\lambda = (\sqrt{\max(\kappa^{\mathrm{w}}_{n_1}, \kappa^{\mathrm{q}}_{n_2})\kappa^{\mathrm{p}}_{n_3}})^{-1}$ refers to the regularization parameter. To find the optimal solution to the general formula, we utilize an inexact augmented lagrange multiplier (IALM) with a proven convergence [21], [42]. The solution procedure of the generalized formula is demonstrated in Algorithm 1. Algorithm 2 illustrates the whole procedure of 3DSS-ILRQ and 3DSSS-ILRQ.

E. Complexity Analysis

Corresponding to the steps in Algorithm 2, 3DSS-ILRQ first costs $\mathcal{O}(rc)$ in randomly selecting $\boldsymbol{p}_{\alpha,\beta}$ to construct the spectral-controlled graphical matrix $\boldsymbol{M}_{\alpha,\beta,\sim}$. Then, the superunit recognition on $\boldsymbol{M}_{\alpha,\beta,\sim}$ costs $\mathcal{O}(blog(b))$ in acquiring (5). The complexity of dividing based on the recognition outcomes to obtain $\{\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3} | n_3 = 1, \ldots, s^{\mathrm{p}}\}$ and $\{\mathrm{X}^{\mathrm{p}}_{n_3} \in \mathbb{R}^{r\times c\times\kappa^{\mathrm{p}}_{n_3}} | n_3 = 1, \ldots, s^{\mathrm{p}}\}$ is $\mathcal{O}(s^{\mathrm{p}})$. Let the iteration times of Algorithm 1 on any subcube $\mathrm{X}^{\mathrm{p}}_{n_3}$ from the 3DSS representation be $T^{\mathrm{p}}_{n_3}$ and on any subcube $\mathrm{X}^{\mathrm{wqp}}_{n_1n_2n_3}$ of the 3DSSS representation be $T^{\mathrm{wqp}}_{n_1n_2n_3}$. The generation of $\tilde{\boldsymbol{L}}^{\mathrm{p}}_{n_3} \in \mathbb{Q}^{r\times c}$ costs the complexity of $\mathcal{O}(r^2c + r^3\kappa^{\mathrm{p}}_{n_3}T^{\mathrm{p}}_{n_3})$ at most. Therefore, the whole ILRQ on $\{\mathrm{X}^{\mathrm{p}}_{n_3} | n_3 = 1, \ldots, s^{\mathrm{p}}\}$ costs $\mathcal{O}\left(s^{\mathrm{p}}r^2c + \sum_{n_3=1}^{s^{\mathrm{p}}} r^3\kappa^{\mathrm{p}}_{n_3}T^{\mathrm{p}}_{n_3}\right)$. In a nutshell, 3DSS-ILRQ costs $\mathcal{O}\left(blog(b) + s^{\mathrm{p}}r^2c + \sum_{n_3=1}^{s^{\mathrm{p}}} r^3\kappa^{\mathrm{p}}_{n_3}T^{\mathrm{p}}_{n_3}\right)$. As shown in Algorithm 2, the complexity of the proposed 3DSSS-ILRQ model is $\mathcal{O}\left(rc + cb + rb + \sum_{n_1n_2n_3=1}^{s^{\mathrm{w}}s^{\mathrm{q}}s^{\mathrm{p}}}((\kappa^{\mathrm{w}}_{n_1})^2\kappa^{\mathrm{q}}_{n_2} + (\kappa^{\mathrm{w}}_{n_1})^3\kappa^{\mathrm{p}}_{n_3}T^{\mathrm{wqp}}_{n_1n_2n_3})\right)$.

IV. Experiments

For a fair comparison, all experiments were performed on a laptop with a 2.60 GHz CPU running on a Windows 11 system, Intel Core i9-13900H processors, and 32 GB of memory.

A. Simulated Hyperspectral Datasets

1) Original HSIs: To quantitatively verify the performance across different types of noise, we conducted simulated experiments using three real-world HSIs.

1) *Indian Pines (IPs):* Acquired by AVIRIS over Northwestern Indiana, USA, on June 12, 1992, the IP dataset contains 200 spectral bands and 145 × 145 pixels, with 10 249 labeled samples divided into 16 classes. It features mixed agricultural and forested areas, including crops like corn and soybeans, with a spatial resolution of ~20 m [43].
2) *Salinas:* Collected by the AVIRIS sensor in the Salinas Valley, California, USA, this dataset includes 204 spectral bands and 512 × 217 pixels, with 54 129 labeled samples across 16 classes (e.g., vegetables, vineyards). It focuses on agricultural fields with a spatial resolution of ~3.7 m, making it ideal for precision agriculture and crop classification [44].
3) *WHU-Hi-Longkou (WHU-LK):* Captured on July 17, 2018, in Longkou, Hubei, China, using a UAV-mounted Headwall Nano-Hyperspec sensor, the WHU-LK dataset contains 270 spectral bands and 550 × 400 pixels, with 204 542 labeled samples covering nine crop species (e.g., corn, rice). Collected at 500-m altitude under clear conditions, it offers a high spatial resolution of 0.463 m [45], [46].

2) Simulated HSI Datasets: The gray values of each HSI are normalized to fall within the range of [0, 1] before further processing. Real HSIs may be corrupted by various types of noise, such as Gaussian noise, impulse noise, deadlines, and stripes. Therefore, five different noises were separately added to the normalized HSIs: Cases 1–5. The ground-truth maps and false-color composite images of the simulated HSI datasets are displayed in Fig. 4.

1) *Case 1 (i.i.d. Gaussian Noise):* All the bands of **H** are contaminated by independent and identically distributed (i.i.d.) Gaussian noise $N(0, \sigma^2)$ with $\sigma = 0.05$. This simulates homogeneous noise across all spectral bands.
2) *Case 2 (Non-i.i.d. Gaussian Noise):* The spectral bands are corrupted by non-i.i.d. Gaussian noise$N(0, \sigma_i^2)$, where σ_i varies across bands. To ensure a fair comparison, the signal-to-noise ratio (SNR) for each band follows a uniform distribution in the range [10, 20] dB. This simulates heterogeneous noise typical in real HSIs.
3) *Case 3 (Non-i.i.d. Gaussian Noise + Stripe Noise):* The non-i.i.d. Gaussian noise setup is consistent with Case 2. In addition, 40 bands are randomly selected to be affected by stripe noise. The stripe noise is generated by adding vertical lines with intensities uniformly distributed in the range [20, 40] % of the maximum pixel value. This simulates artifacts caused by sensor malfunctions or calibration errors.
4) *Case 4 (Non-i.i.d. Gaussian Noise + Impulse Noise):* The non-i.i.d. Gaussian noise setup is identical to Case 2. Impulse noise is introduced to 40 randomly selected spectral channels, with the percentage of corrupted

$$\tilde{x}^{n_3}_{\alpha,\beta} = \begin{cases} h_{\alpha,\beta,\theta^{\mathrm{p}}_{n_3}}, & \kappa^{\mathrm{p}}_{n_3} = 1 \\ h_{\alpha,\beta,(\theta^{\mathrm{p}}_{n_3}-1)} + h_{\alpha,\beta,\theta^{\mathrm{p}}_{n_3}}u, & \kappa^{\mathrm{p}}_{n_3} = 2 \\ h_{\alpha,\beta,(\theta^{\mathrm{p}}_{n_3}-2)} + h_{\alpha,\beta,(\theta^{\mathrm{p}}_{n_3}-1)}u + h_{\alpha,\beta,\theta^{\mathrm{p}}_{n_3}}v, & \kappa^{\mathrm{p}}_{n_3} = 3 \\ h_{\alpha,\beta,\left(\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3}\right)^{0.1}} + h_{\alpha,\beta,\left(\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3}\right)^{0.3}}u + h_{\alpha,\beta,\left(\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3}\right)^{0.6}}v + h_{\alpha,\beta,\left(\boldsymbol{\Gamma}^{\mathrm{p}}_{n_3}\right)^{0.9}}\delta, & \kappa^{\mathrm{p}}_{n_3} \geq 4 \end{cases}$$

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Algorithm 1 Solve (12) by IALM

Input : $\tilde{\boldsymbol{X}} = (\tilde{x}_{\alpha,\beta})_{\tau_1\times\tau_2}$, τ_3, $\lambda = (\sqrt{\max(\tau_1,\tau_2)}\tau_3)^{-1}$, $\rho = 1.5$, $\varepsilon = 10^{-7}$

Output: low-rank $\tilde{\boldsymbol{L}}$

Initialize : $\mu_0 = 1.25/\max(\sigma(\tilde{\boldsymbol{X}}))$, $\tilde{\boldsymbol{N}}_0 = \tilde{\boldsymbol{0}} \in \mathbb{Q}^{\tau_1\times\tau_2}$, $k = 0$ $\tilde{\boldsymbol{Y}}_0 = \tilde{\boldsymbol{X}}/\max(\max(\sigma(\tilde{\boldsymbol{X}})), \lambda^{-1}\max|\tilde{x}_{\alpha,\beta}|)$

While not converged, **do**

1: Update $\tilde{\boldsymbol{L}}_{k+1}$ by

$$\tilde{\boldsymbol{L}}_{k+1} = \arg\min_{\tilde{\boldsymbol{L}}} \left(\left\|\tilde{\boldsymbol{L}}\right\|_{\dagger} + \frac{\mu_k}{2} \left\| \tilde{\boldsymbol{L}} + \tilde{\boldsymbol{N}}_k - \tilde{\boldsymbol{X}} + \frac{\tilde{\boldsymbol{Y}}_k}{\mu_k} \right\|_{\mathrm{F}} \right)$$

2 : Update $\tilde{\boldsymbol{N}}_{k+1}$ by

$$\tilde{\boldsymbol{N}}_{k+1} = \arg\min_{\tilde{\boldsymbol{N}}} \left(\left\|\tilde{\boldsymbol{N}}\right\|_{2,1} + \frac{\mu_k}{2} \left\| \tilde{\boldsymbol{L}}_{k+1} + \tilde{\boldsymbol{N}} - \tilde{\boldsymbol{X}} + \frac{\tilde{\boldsymbol{Y}}_k}{\mu_k} \right\|_{\mathrm{F}} \right)$$

3 : $\tilde{\boldsymbol{Y}}_{k+1} = \tilde{\boldsymbol{Y}}_k + \mu_k(\tilde{\boldsymbol{X}} - \tilde{\boldsymbol{L}}_{k+1} - \tilde{\boldsymbol{N}}_{k+1})$

4 : $\mu_{k+1} = \rho\mu_k$

5 : $k = k + 1$

Check the convergence conditions

$$\frac{\left\|\tilde{\boldsymbol{X}} - \tilde{\boldsymbol{L}}_k - \tilde{\boldsymbol{N}}_k\right\|_{\mathrm{F}}}{\left\|\tilde{\boldsymbol{X}}\right\|_{\mathrm{F}}} < \varepsilon$$

End while

Algorithm 2 3DSS-ILRQ [*3DSSS-ILRQ*]

Input : $\mathbf{H} \in \mathbb{R}^{r\times c\times b}$

Output : low-rank $\mathbf{L} \in \mathbb{R}^{r\times c\times b}$

1 : Acquire $\mathbf{M}_{\alpha,\beta,\sim}$ using (2)
[+acquire $\mathbf{M}_{\sim,\beta,\phi}$ using (3) and $\mathbf{M}_{\alpha,\sim,\phi}$ using (4)]

2 : Perform (5) on $\mathbf{M}_{\alpha,\beta,\sim}$ to generate $\mathbf{S}^{\mathrm{p}}_{n_3}$ as (6)
[+perform (5) on $\mathbf{M}_{\sim,\beta,\phi}$ and $\mathbf{M}_{\alpha,\sim,\phi}$ to generate $\mathbf{S}^{\mathrm{w}}_{n_1}$ and $\mathbf{S}^{\mathrm{q}}_{n_2}$ like (6), respectively]

3 : Obtain $\mathbf{\Gamma}^{\mathrm{p}}_{n_3}$ in (7) using $\mathbf{S}^{\mathrm{p}}_{n_3}(n_3 = 1,\ldots,s^{\mathrm{p}})$
[+obtain $\mathbf{\Gamma}^{\mathrm{w}}_{n_1}(n_1 = 1,\ldots,s^{\mathrm{w}})$ and $\mathbf{\Gamma}^{\mathrm{q}}_{n_2}(n_2 = 1,\ldots,s^{\mathrm{q}})$ in the same way]

4 : Obtain $\mathrm{X}^{\mathrm{p}}_{n_3}(n_3 = 1,\ldots,s^{\mathrm{p}})$ using (8)
[obtain $\mathrm{X}^{\mathrm{wqp}}_{n_1n_2n_3}(n_1 = 1,\ldots,s^{\mathrm{w}}; n_2 = 1,\ldots,s^{\mathrm{q}}; n_3 = 1,\ldots,s^{\mathrm{p}})$ by (9)]

For $n_3 = 1,\ldots,s^{\mathrm{p}}$ **do**[**for** $n_1 = 1,\ldots,s^{\mathrm{w}}$ **do**;**for** $n_2 = 1,\ldots,s^{\mathrm{q}}$ **do**]

5 : Transform $\mathrm{X}^{\mathrm{p}}_{n_3} \in \mathbb{R}^{r\times c\times\kappa^{\mathrm{p}}_{n_3}}$ to $\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3} \in \mathbb{Q}^{r\times c}$ using (10)
[transform $\mathrm{X}^{\mathrm{wqp}}_{n_1n_2n_3} \in \mathbb{R}^{\kappa^{\mathrm{w}}_{n_1}\times\kappa^{\mathrm{q}}_{n_2}\times\kappa^{\mathrm{p}}_{n_3}}$ to $\tilde{\boldsymbol{X}}^{\mathrm{wqp}}_{n_1n_2n_3} \in \mathbb{Q}^{\kappa^{\mathrm{w}}_{n_1}\times\kappa^{\mathrm{q}}_{n_2}}$]

6 : Perform Algorithm 1 on $\tilde{\boldsymbol{X}}^{\mathrm{p}}_{n_3}$ in (11) with adaptive $\lambda^{\mathrm{p}}_{n_3}$ to acquire $\tilde{\boldsymbol{L}}^{\mathrm{p}}_{n_3}$
[perform Algorithm 1 on $\mathrm{X}^{\mathrm{wqp}}_{n_1n_2n_3}$ in (12) with adaptive λ to acquire $\tilde{\boldsymbol{L}}^{\mathrm{wqp}}_{n_1n_2n_3}$]

7 : Transform $\tilde{\boldsymbol{L}}^{\mathrm{p}}_{n_3} \in \mathbb{Q}^{r\times c}$ to $\mathbf{L}^{\mathrm{p}}_{n_3} \in \mathbb{R}^{r\times c\times\kappa^{\mathrm{p}}_{n_3}}$ using the inverse transformation of (10)
[transform $\tilde{\boldsymbol{L}}^{\mathrm{wqp}}_{n_1n_2n_3} \in \mathbb{Q}^{\kappa^{\mathrm{w}}_{n_1}\times\kappa^{\mathrm{q}}_{n_2}}$ to $\mathbf{L}^{\mathrm{wqp}}_{n_1n_2n_3} \in \mathbb{R}^{\kappa^{\mathrm{w}}_{n_1}\times\kappa^{\mathrm{q}}_{n_2}\times\kappa^{\mathrm{p}}_{n_3}}$ using the inverse processing of step 5]

End [**end**;**end**]

8 : Compute the inverse function of (8) on $\mathbf{L}^{\mathrm{p}}_{n_3}(n_3 = 1,\ldots,s^{\mathrm{p}})$ to acquire $\mathbf{L} \in \mathbb{R}^{r\times c\times b}$ in (1)
[compute the inverse function of (9) on $\mathbf{L}^{\mathrm{wqp}}_{n_1n_2n_3}(n_1 = 1,\ldots,s^{\mathrm{w}}; n_2 = 1,\ldots,s^{\mathrm{q}}; n_3 = 1,\ldots,s^{\mathrm{p}})$ to acquire $\mathbf{L} \in \mathbb{R}^{r\times c\times b}$]

pixels ranging from 50% to 70%. The impulse noise is modeled as random pixel replacements with either the minimum (0) or maximum (1) normalized values, simulating salt-and-pepper noise.

5) *Case 5 (Non-i.i.d. Gaussian Noise + Deadline Noise):* The non-i.i.d. Gaussian noise setup is the same as in Case 2. Deadline noise is added to 50 randomly chosen bands, where each deadline is a continuous line of corrupted pixels with intensities set to zero. The number of deadlines per band ranges from 5 to 15, simulating sensor defects or data transmission errors.

B. Experimental Settings

1) Comparison Methods: We consider various low-rank-related manners including: 1) matrix LRR: non-i.i.d. mixture of Gaussians-targeted low-rank factorization (NMOG-LRF) modeling the matrixed noise under the Bayesian framework [10]; 2) tensor LRR: multilinear Tucker decomposition-based LRR (MTDL) [13], Dual ℓp quasinorm and weighted Schatten p-norm are introduced to restrict the sparse errors of TRPCA (DP-TRPCA) [14], and new nuclear norm TRPCA (NT-TRPCA), designing a novel regularization norm with t-product to ensure exact recovery [18]; 3) hypercomplex LRR: polar n-complex singular value decomposition (PN-CSVD) that extracts ideal features by hypercomplex and tensor-based singular value decomposition [21]; and 4) deep-learning LRR: hyperspectral blind reconstruction from spectral subsampling (HY-D) using a plug-and-play network to exploit spectral low-rankness and spatial correlations [23] and the S^{2}Mamba model, which depends on a selective structured state space model to efficiently capture long-range dependencies with linear complexity [47]. It is worth noting that HY-D and S^{2}Mamba are deep-learning models implemented in Python with pretrained weights, while other comparison methods are optimization-based approaches implemented in MATLAB. This fundamental difference in computational paradigms and implementation platforms makes direct runtime comparisons not technically meaningful, though all methods are evaluated fairly using identical training and testing sets.

2) Training Setup: In light of the extensive global spatial samples available, we chose to use 10% of labeled pixels from each class in the IP dataset, 2% of labeled pixels from each class in the Salinas dataset, and 5‰ randomly selected labeled pixels from each class in WHU-LK to construct training sets for effective testing and comparison. Using smaller ratios for larger datasets simulates scenarios where labeled samples are scarce, thereby testing the robustness and generalization ability of the methods. The remaining portion of each dataset was designated as the testing set, comprising the remaining labeled pixels. The classification experiments underwent five-fold cross-validation for result averaging to ensure robust and consistent results.

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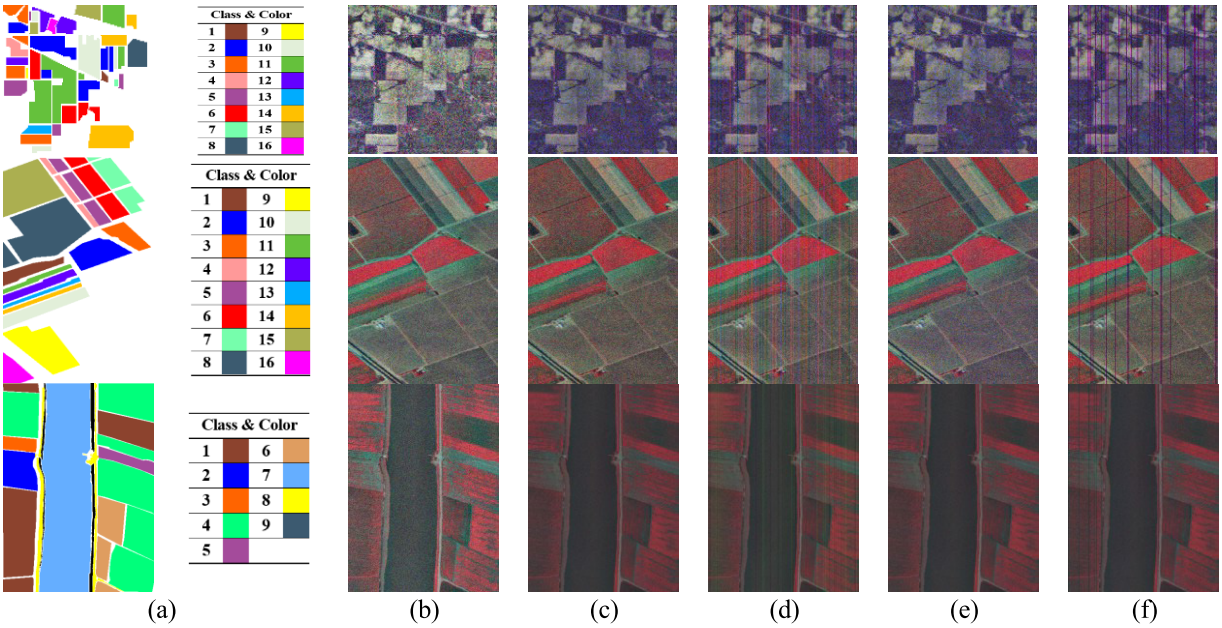


Fig. 4. IP (first row), Salinas (second row), and WHU-LK (third row). (a) Ground-truth map (white pixels indicate unlabeled samples) and false-color image with R-G-B of 22–28-11 bands on IP, 45-28-14 bands on Salinas, and 212-80-66 bands on WHU-LK under (b) Case 1, (c) Case 2, (d) Case 3, (e) Case 4, and (f) Case 5.

TABLE I

CLASSIFICATION ACCURACIES (%) WITH (±) STD UNDER VARIOUS NOISY CASES ON IP (10% LABELED SAMPLES)

Cases	Metrics	NMOG-LRF	MTDL	DP-TRPCA	NT-TRPCA	PN-CSVD	HY-D	S^2Mamba	3DSS-ILRQ	3DSSS-ILRQ
Case 1	M-OA	56.35±0.41	70.48±0.79	39.43±0.73	51.85±0.46	48.68±0.47	72.13±1.36	*81.85±1.04*	**87.75±0.57**	78.75±0.03
	M-AA	50.40±2.49	74.82±0.98	16.45±2.98	41.92±6.48	26.63±3.46	66.27±1.55	81.74±2.49	*83.63±3.20*	**86.25±0.03**
	M-Kappa	48.83±0.32	66.09±0.95	23.57±1.17	41.72±0.58	37.36±0.72	68.13±1.56	79.25±1.20	**85.95±0.66**	*75.48±0.04*
Case 2	M-OA	80.17±1.18	59.32±0.95	55.08±0.67	70.85±0.27	55.55±0.71	74.13±1.15	84.17±0.87	**93.27±0.38**	*87.03±0.08*
	M-AA	80.87±2.46	53.86±1.89	47.34±1.88	64.98±4.52	48.15±3.77	69.39±2.11	82.70±2.13	**91.09±1.86**	*89.35±0.12*
	M-Kappa	77.30±1.36	52.00±1.11	46.42±0.84	66.32±0.29	47.12±0.89	70.41±1.35	81.91±0.96	**92.31±0.44**	*85.15±0.10*
Case 3	M-OA	60.59±0.75	62.60±1.07	56.81±0.78	74.96±0.80	57.82±0.94	71.83±1.11	*86.92±0.53*	**86.93±1.17**	83.30±0.05
	M-AA	55.35±4.62	59.98±1.00	52.14±5.22	79.93±3.84	53.44±3.84	69.10±0.90	*84.09±1.42*	82.91±3.19	**87.92±0.12**
	M-Kappa	54.17±0.98	55.81±1.27	48.29±1.02	71.02±0.93	49.78±1.21	67.80±1.26	**85.11±0.60**	*85.03±1.36*	80.81±0.06
Case 4	M-OA	58.10±0.85	57.35±0.75	53.71±0.46	71.81±0.50	53.46±0.63	67.06±0.76	83.89±0.46	**92.17±0.67**	*85.37±0.08*
	M-AA	50.62±3.52	51.58±2.33	45.59±3.57	66.04±1.97	44.53±3.73	60.88±2.28	82.43±1.80	**90.18±1.62**	*88.51±0.20*
	M-Kappa	50.84±1.23	49.35±0.86	44.39±0.74	67.48±0.56	44.15±0.91	62.26±0.90	81.59±0.55	**91.05±0.77**	*83.23±0.09*
Case 5	M-OA	57.65±0.53	61.00±0.55	54.47±0.75	73.03±0.59	55.32±0.91	64.89±0.81	**85.41±0.63**	*85.33±0.23*	82.60±0.07
	M-AA	48.44±1.48	59.15±0.59	46.76±3.53	73.78±3.13	48.23±6.37	61.24±2.27	*83.49±2.04*	83.27±3.15	**86.55±0.08**
	M-Kappa	50.49±0.78	53.76±0.68	45.10±1.06	68.66±0.68	46.29±1.26	59.45±1.05	*82.93±0.71*	**83.16±0.27**	*80.00±0.08*

C. Evaluation of Numerical Results

In this section, we utilize a multiclass support vector machine (MSVM) classifier to evaluate the effectiveness of LRR methods in classification tasks. The MSVM classifier is applied to all testing datasets using a radial basis function kernel with fixed parameters $c = 10$ and $\gamma = 0.125$ [48]. To ensure a fair comparison, consistent parameter settings are maintained for the MSVM classifier, and each cross-validation on the same image is predetermined rather than random. We include three evaluation metrics to assess the detection performance quantitatively: overall accuracy (OA), average accuracy of each category (AA), and kappa coefficient (Kappa). The fivefold cross-validation results in mean OA (M-OA), mean averaging accuracy (M-AA), and mean kappa (M-Kappa) with corresponding standard deviations (std). The best results are highlighted in bold, and the suboptimal values are marked in underlined italics.

The selection of distinct evaluation metrics provides a comprehensive assessment of classification performance. While M-OA measures global correctness, it can be biased by class imbalance, favoring dominant classes. M-AA addresses this limitation by evaluating the mean accuracy across all classes, ensuring balanced performance for smaller or underrepresented classes. M-Kappa, on the other hand, accounts for random chance, making it robust to class imbalance. Discrepancies in performance across these metrics often arise due to class imbalance, noise effects, or regional directionality.

1) Indian Pine: The results presented in Table I display the overall performance of various algorithms. The proposed 3DSS-ILRQ consistently achieves the highest M-OA and M-Kappa across most noisy cases, demonstrating its

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TABLE II
CLASSIFICATION ACCURACIES (%) WITH (±) STD UNDER VARIOUS NOISY CASES ON SALINAS (2% LABELED SAMPLES)

Cases	Metrics	NMOG-LRF	MTDL	DP-TRPCA	NT-TRPCA	PN-CSVD	HY-D	S^2Mamba	3DSS-ILRQ	3DSSS-ILRQ
Case 1	M-OA	77.51±0.37	83.36±0.70	48.72±0.35	75.87±0.51	60.45±0.17	*89.48±0.24*	88.79±0.55	**91.91±0.03**	88.75±0.07
	M-AA	80.61±0.21	87.11±0.31	36.42±0.56	83.41±0.65	63.59±0.55	**91.65±0.56**	87.67±0.69	85.57±0.07	*87.95±0.09*
	M-Kappa	74.83±0.43	81.40±0.79	39.71±0.44	72.84±0.58	54.49±0.23	*88.28±0.27*	87.49±0.46	**90.99±0.04**	87.46±0.08
Case 2	M-OA	83.30±0.54	80.87±0.37	79.05±0.47	86.93±0.37	79.36±0.57	86.42±0.39	*94.81±0.12*	**96.48±0.04**	93.13±0.03
	M-AA	88.45±0.60	87.02±0.30	86.47±0.19	91.83±0.22	86.56±0.16	89.85±0.34	**96.79±0.06**	94.00±0.09	*94.55±0.03*
	M-Kappa	81.35±0.59	78.62±0.41	76.50±0.54	85.42±0.40	76.86±0.69	84.87±0.44	*94.21±0.32*	**96.08±0.04**	92.36±0.04
Case 3	M-OA	83.18±0.18	80.63±0.32	78.23±0.66	87.03±0.45	79.56±0.23	86.94±0.30	*92.63±0.14*	**94.28±0.05**	90.09±0.04
	M-AA	88.18±0.24	87.20±0.34	85.12±2.61	**91.61±0.22**	86.72±0.04	88.91±0.14	90.19±0.98	90.36±0.10	*90.43±0.04*
	M-Kappa	81.20±0.21	78.32±0.37	75.51±0.82	85.55±0.51	77.09±0.27	85.45±0.33	*91.78±0.47*	**93.63±0.05**	88.97±0.04
Case 4	M-OA	82.72±0.53	79.22±0.46	77.27±0.73	87.58±0.18	77.54±0.62	88.78±0.29	90.88±0.14	**96.07±0.02**	*93.16±0.04*
	M-AA	87.55±0.69	85.85±0.29	85.42±0.23	92.41±0.25	85.26±0.39	89.87±0.56	89.84±1.07	*93.12±0.07*	**94.35±0.03**
	M-Kappa	80.70±0.58	76.71±0.57	74.44±0.88	86.14±0.20	74.79±0.71	87.51±0.32	89.83±0.44	**95.62±0.03**	*92.39±0.04*
Case 5	M-OA	82.79±0.48	79.42±0.24	77.21±0.61	86.09±0.65	78.00±0.55	85.12±0.43	*91.53±0.23*	**91.74±0.05**	89.87±0.05
	M-AA	88.08±0.32	86.15±0.15	85.62±0.50	**91.02±0.33**	85.70±0.41	87.72±0.25	*90.32±0.06*	85.90±0.07	90.14±0.04
	M-Kappa	80.75±0.54	76.97±0.27	74.38±0.70	84.49±0.74	75.31±0.63	83.40±0.51	*90.69±0.71*	**90.79±0.06**	88.71±0.06

TABLE III
CLASSIFICATION ACCURACIES (%) WITH (±) STD UNDER VARIOUS NOISY CASES ON WHU-LK (5‰ LABELED SAMPLES)

Cases	Metrics	NMOG-LRMF	MTDL	DP-TRPCA	NT-TRPCA	PN-CSVD	HY-D	S^2Mamba	3DSS-ILRQ	3DSSS-ILRQ
Case 1	M-OA	88.46±0.22	92.46±0.31	81.04±0.46	90.59±0.07	86.54±0.22	89.38±0.24	91.14±0.86	*94.33±0.49*	**94.47±0.03**
	M-AA	73.52±6.10	82.51±2.21	55.00±0.85	81.95±0.76	64.59±5.97	72.97±1.99	**91.52±0.09**	*88.56±1.09*	86.42±0.12
	M-Kappa	84.60±0.34	90.03±0.39	73.68±0.68	87.46±0.11	81.86±0.32	85.90±0.29	88.08±1.25	*92.52±0.65*	**92.70±0.04**
Case 2	M-OA	91.05±0.77	90.61±0.11	89.02±0.28	93.66±0.23	89.43±0.40	87.81±0.27	93.38±0.35	*95.14±0.20*	**95.86±0.04**
	M-AA	81.24±0.90	81.74±0.94	77.67±6.37	86.66±1.32	76.76±6.23	65.52±4.13	**93.81±0.44**	89.68±0.86	*90.09±0.06*
	M-Kappa	88.13±1.05	87.48±0.14	85.31±0.38	91.61±0.31	85.89±0.54	83.70±0.39	91.15±0.50	*93.59±0.26*	**94.54±0.05**
Case 3	M-OA	89.56±0.64	90.52±0.23	88.84±0.26	*93.57±0.30*	89.39±0.39	87.64±0.25	93.37±0.31	93.18±0.41	**95.24±0.02**
	M-AA	76.20±2.54	81.66±0.98	78.98±6.90	86.65±1.51	77.45±6.28	64.04±5.93	**94.12±0.06**	84.67±1.29	*88.70±0.07*
	M-Kappa	86.13±0.89	87.36±0.30	85.05±0.35	*91.49±0.40*	85.82±0.53	83.45±0.35	91.13±0.28	91.01±0.55	**93.73±0.03**
Case 4	M-OA	89.89±0.25	89.16±0.24	88.64±0.16	93.80±0.22	88.62±0.23	87.54±0.27	92.53±0.15	*95.18±0.39*	**95.23±0.02**
	M-AA	76.85±2.03	78.51±2.03	78.52±3.68	86.72±1.13	79.43±0.97	63.06±3.94	**92.69±0.09**	*89.98±1.04*	88.39±0.07
	M-Kappa	86.60±0.30	85.49±0.33	84.77±0.22	91.80±0.29	84.75±0.32	83.30±0.37	89.98±0.77	*93.64±0.52*	**93.71±0.03**
Case 5	M-OA	89.37±0.49	90.34±0.16	88.94±0.14	93.57±0.23	89.53±0.16	87.62±0.22	92.97±0.15	*93.67±0.70*	**95.17±0.03**
	M-AA	74.82±3.23	81.29±0.75	82.20±1.06	*87.10±1.32*	81.47±0.72	66.41±3.65	**93.82±0.41**	86.70±1.16	*88.77±0.11*
	M-Kappa	85.87±0.68	87.10±0.21	85.18±0.18	91.50±0.30	86.01±0.22	83.42±0.29	90.57±0.21	*91.65±0.94*	**93.63±0.04**

robustness. However, 3DSSS-ILRQ outperforms 3DSS-ILRQ in M-AA under Case 1, Case 3, and Case 5, with improvements of 2.62%, 5.01%, and 3.28%, respectively. This complementary performance suggests that the two proposed algorithms excel in different aspects, with 3DSS-ILRQ being more effective in OA and 3DSSS-ILRQ in average accuracy under specific noise conditions. Other algorithms exhibit varying unstable adaptability to different noise scenarios. S^2Mamba typically ranks behind the two proposed methods while remaining comparable to strong baselines such as NT-TRPCA and HY-D, particularly in Cases 3 and Case 5.

2) Salinas: The averaging results of all algorithms under various noisy simulated cases on the Salinas dataset are shown in Table II. The proposed 3DSSS-ILRQ and 3DSS-ILRQ dominate M-OA, M-AA, and M-Kappa across most scenarios. HY-D achieves the highest M-AA in Case 2, and NT-TRPCA outperforms in Cases 3 and 5, indicating that these methods are better suited for specific noise types. Notably, HY-D ranks second in Case 1 for M-OA and M-Kappa, showcasing its adaptability to i.i.d. Gaussian noise. NT-TRPCA ranks second on M-AA in Cases 3 and 5, indicating its robustness to complex noise. S^2Mamba achieves remarkable performance with M-OA ranging, particularly excelling in Case 2 and demonstrating superior noise adaptability. Nevertheless, 3DSS-ILRQ generally outperforms these methods. A key observation is that 3DSS-ILRQ exhibits slightly lower robustness in M-AA compared to M-OA and M-Kappa, possibly due to its reliance on regional directionality. This limitation is more pronounced in Salinas, where the data distribution is more complex and less structured than IP and WHU-LK.

3) WHU-LK: The overall accuracies of all algorithms facing various noisy simulated cases on WHU-LK are shown in Table III. The proposed 3DSSS-ILRQ achieves the highest M-OA, M-AA, and M-Kappa in most cases, except for M-AA in all cases, where S^2Mamba performs slightly better. Compared with other datasets, S^2Mamba demonstrates better performance on WHU-LK. This difference may be attributed to the higher data quality of WHU-LK, which makes it less susceptible to noise-induced degradation for deep-learning models. Despite this, the proposed 3DSSS-ILRQ and 3DSS-ILRQ maintain superior and robust performance across all simulated noisy situations, even with limited labeled samples. The proposed methods' ability to adaptively identify and process homogeneous regions (spatially and spectrally) makes them highly effective in diverse noise scenarios. Therefore, 3DSS-ILRQ and 3DSSS-ILRQ consistently achieve superior

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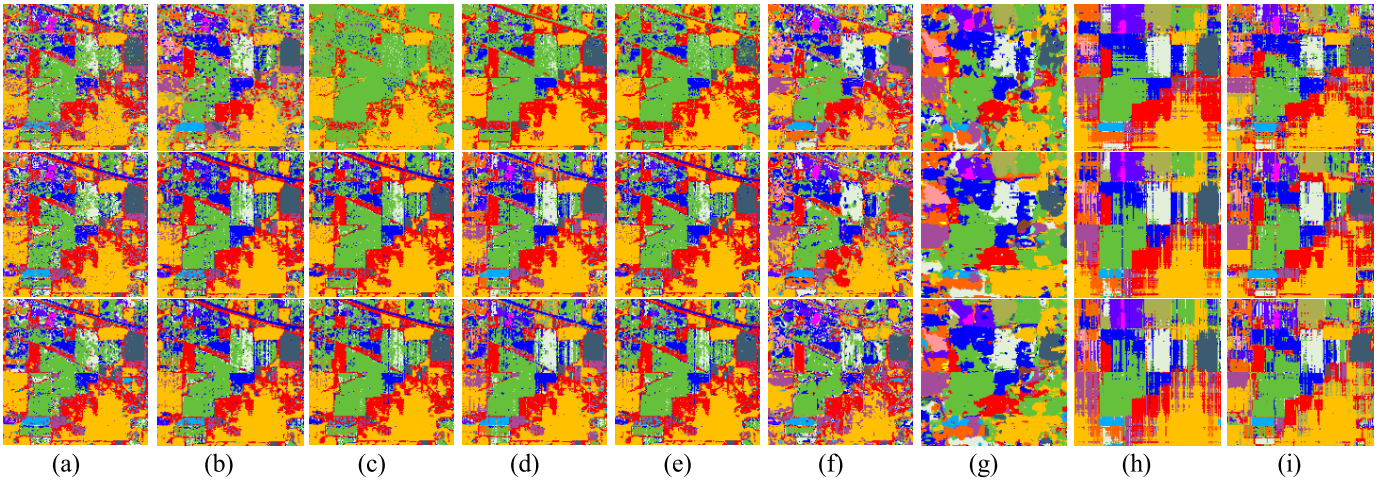


Fig. 5. Classification maps of different algorithms on IP under Case 1 (first row), Case 3 (second row), and Case 5 (third row). (a) NMOG-LRMF, (b) MTDL, (c) DP-TRPCA, (d) NT-TRPCA, (e) PN-CSVD, (f) HY-D, (g) S^2Mamba, (h) 3DSS-ILRQ, and (i) 3DSSS-ILRQ.

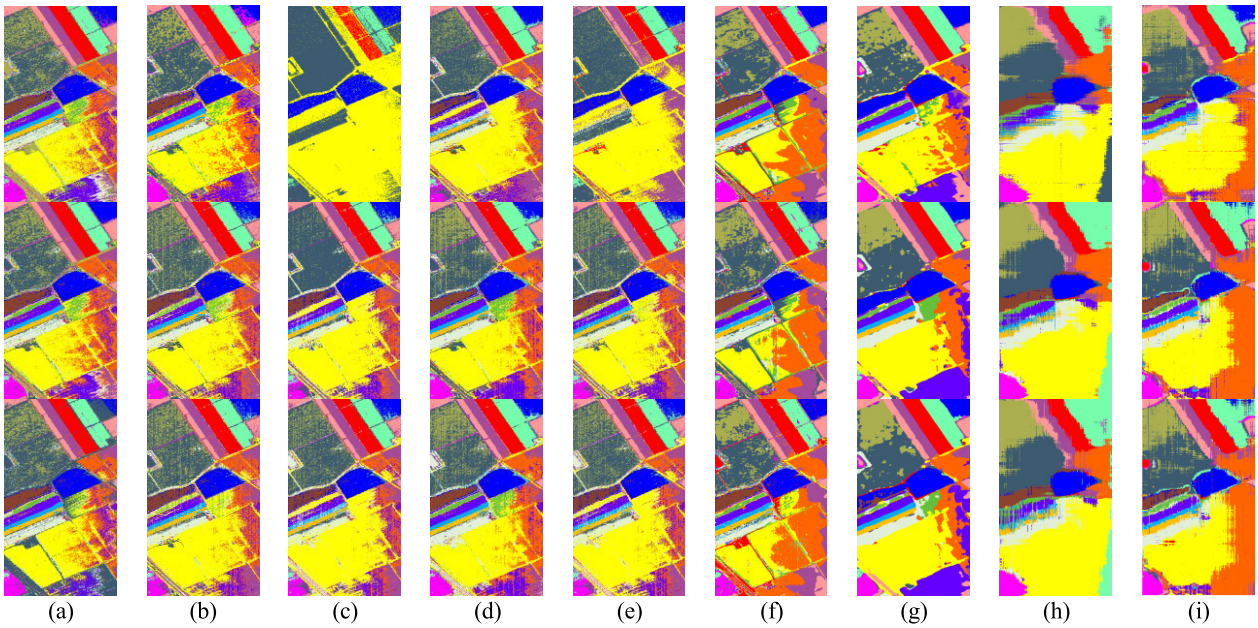


Fig. 6. Classification maps of different algorithms on Salinas under Case 1 (first row), Case 3 (second row), and Case 5 (third row). (a) NMOG-LRMF, (b) MTDL, (c) DP-TRPCA, (d) NT-TRPCA, (e) PN-CSVD, (f) HY-D, (g) S^2Mamba, (h) 3DSS-ILRQ, and (i) 3DSSS-ILRQ.

performance across all three datasets under various noise scenarios.

D. Evaluation of Visualization Quality

1) Classification Maps: Figs. 5–7 display the classification maps generated by the MSVM classifier under Case 1, Case 3, and Case 5. It is worth noting that, unlike the ground truth map in Fig. 4(a) where unlabeled regions are shown in white, the classification maps deliberately display predictions for all pixels across the entire scene, including those unlabeled regions. This visualization strategy enables a direct visual comparison with the false-color composite images in Fig. 4(b)–(f), providing a comprehensive view of how different algorithms handle noisy data across the complete spatial domain. Across all datasets, it is noticeable that the comparison methods consistently exhibit numerous outliers in areas that should ideally have a uniform color distribution. For instance, in the WHU-LK dataset, the blue spatial points of Class 2 are intermingled with the purple points of Class 5 in Fig. 7, contrary to the ground-truth map.

This discrepancy could be attributed to noise interference affecting the algorithms' ability to differentiate and recognize pixels of distinct categories. However, the results of our proposed 3DSSS-ILRQ and 3DSS-ILRQ algorithms do not exhibit such severe confusion. These algorithms produce global spatial pixels that are not impacted by noise or mixing but are densely clustered by region. The numerical measures of classification support the consistency of these findings. It's important to note that while the proposed algorithms are robust against noisy points in pixel recognition, they may compromise the recognition of edge linear features. While the accuracy in ground object classification remains high, there may be more noticeable errors in target recognition and the detection of particular objects. This limitation

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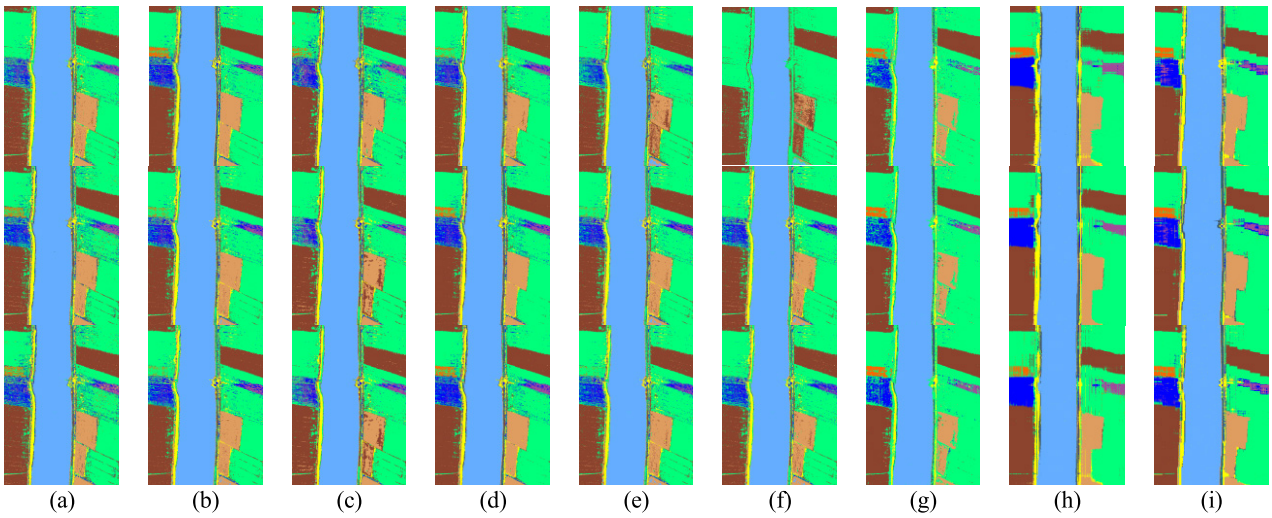


Fig. 7. Classification maps of different algorithms on WHU-LK under Case 1 (first row), Case 3 (second row), and Case 5 (third row). (a) NMOG-LRMF, (b) MTDL, (c) DP-TRPCA, (d) NT-TRPCA, (e) PN-CSVD, (f) HY-D, (g) S^{2}Mamba, (h) 3DSS-ILRQ, and (i) 3DSSS-ILRQ.

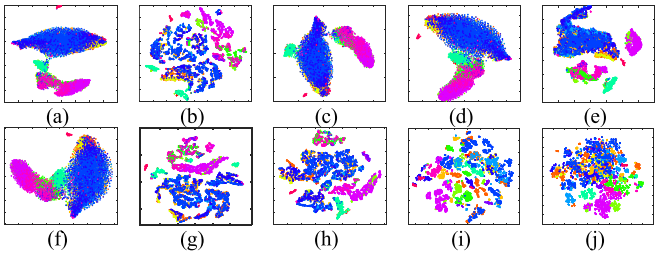


Fig. 8. T-SNE of features obtained by (a) original simulated HSI, (b) NMOG-LRMF, (c) MTDL, (d) DP-TRPCA, (e) NT-TRPCA, (f) PN-CSVD, (g) HY-D, (h) S^{2}Mamba, (i) 3DSS-ILRQ, and (j) 3DSSS-ILRQ on IP under Case 2.

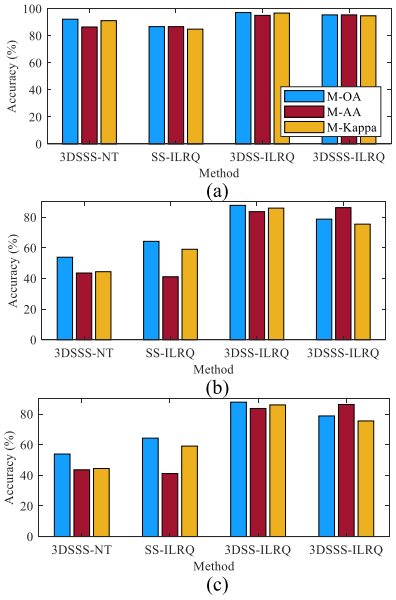


Fig. 9. Performance of 3DSSS-NT-TRPCA (3DSSS-NT), superpixel SS-ILRQ, 3DSS-ILRQ, and 3DSSS-ILRQ on (a) original IP and noisy IP under (b) Case 1 and (c) Case 5.

highlights the need for further research and optimization in the future.

2) Spatial Pixel Separability Visualization: The t-distributed stochastic neighbor embedding (t-SNE) technique

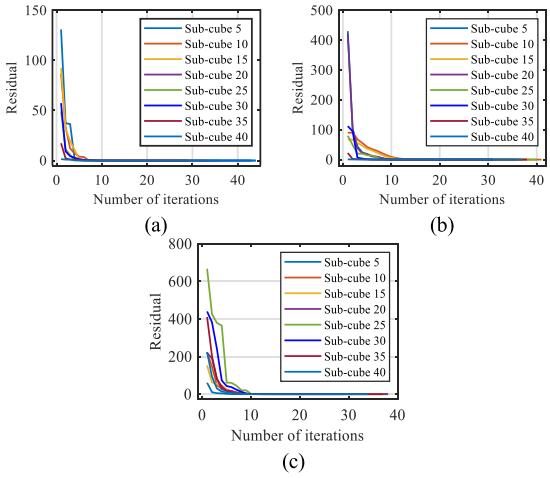


Fig. 10. Convergence curves on selected supercubes on (a) IP, (b) Salinas, and (c) WHU-LK.

functions on differentiating groups of observed data by projecting high-dimensional features into a low-dimensional space [49]. Its effectiveness relies on the input features to create a sample separability graph containing pixels labeled with different categories. Hence, t-SNE visually showcases the influence of various low-rank-related feature techniques on subsequent classification tasks by depicting the separability graph of labeled spatial pixels. Fig. 8 displays the t-SNE graph of the features on labeled pixels obtained by different algorithms in Case 2 derived from IP. As depicted in Fig. 8, the labeled samples of MTDL, DP-TRPCA, NT-TRPCA, and PN-CSVD exhibit severe overlap, where the blue point cluster nearly encompasses all small labeled areas. The results of NMOG-LRMF, HY-D, and S^{2}Mamba show a slight improvement but still exhibit noticeable inseparability and overlap. On the contrary, our models effectively separate compact observation points and reduce the degree of overlap among points with distinct colors. The visualization generally aligns with the classification results, demonstrating the effectiveness of our models in recognizing pixel categories.

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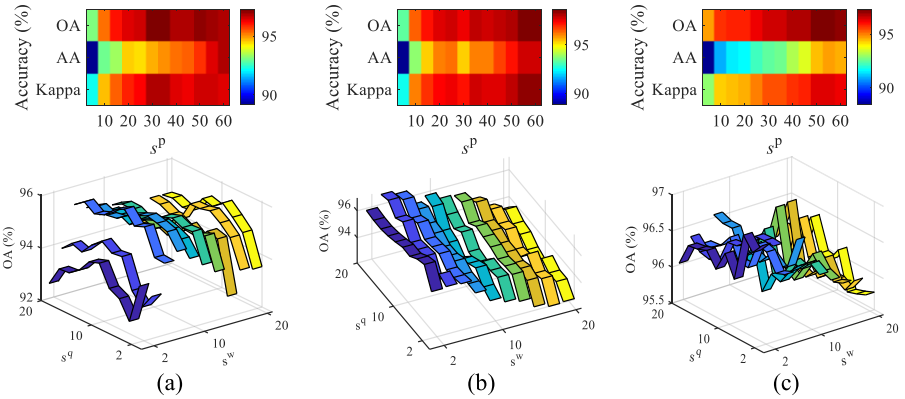


Fig. 11. Performance on different sets of the numbers of superspatial-rows s^w, superspatial-columns s^q, and superspectral units s^p on (a) IP, (b) Salinas, and (c) WHU-LK.

TABLE IV

RUNTIME (S) OF DIFFERENT ALGORITHMS ON IP (FIRST LINE), SALINAS (SECOND LINE), AND WHU-LK (THIRD LINE)

NMOG-LRMF	MTDL	DP-TRPCA	NT-TRPCA	PN-CSVD	3DSS-ILRQ	3DSSS-ILRQ
73.1	403.6	625.7	96.7	167.6	32.4	90.8
131.3	2413.7	1586.4	372.9	289.5	61.6	1126.6
402.2	5526.2	2558.5	1169.2	1807.9	79.6	477.2

TABLE V

M-OA (%) OF S^2 MAMBA AND 3DSS-ILRQ UNDER NOISE INTENSITIES ON IP WITH SVM AND KNN

Cases	S^2Mamba		3DSS-ILRQ	
	SVM	KNN	SVM	KNN
Case 1	78.99	64.35	**85.94**	**78.34**
Case 2	80.47	64.31	**90.05**	**83.20**
Case 4	81.11	65.61	**85.73**	**67.77**

E. Empirical Analysis

In this section, we conduct an ablation study, convergence analysis, runtime comparison, and hyperparameter sensitivity study to explore the properties of the proposed models.

1) Ablation Study: To assess the effectiveness of the critical components of our models (3DSSS and ILRQ), we conducted a comparison with 3DSSS-NT-TRPCA, which is a superpixel segmentation-based ILRQ (SS-ILRQ), using original and Cases 1 and 5 on IP. The overall classification accuracy results are presented in Fig. 9. As depicted in Fig. 9, the proposed 3DSS-ILRQ and 3DSSS-ILRQ significantly outperform the comparative model, especially in noisy cases. This validates the benefit of our 3-D superunit recognition for spectral and spatial heterogeneity. Furthermore, our ILRQ demonstrates greater adaptability in the LRR of HSI.

2) Convergence Display: We have plotted the residual curves of the LRR operating on the fifth, tenth, 15th, 20th, 25th, 30th, 35th, and 40th supercubes obtained by the 3DSS-ILRQ on the IP, Salinas, and WHU-LK datasets under Case 1 (refer to Fig. 10). As depicted in Fig. 10, the superblocks converge in 40 iterations across different datasets. This study sets the maximum number of iterations on each subcube to 100, ensuring the IALM achieves optimal optimization performance for all hyperspectral datasets. Regardless of their initial size, the residual values can decrease rapidly before the tenth iteration and stabilize close to 0 before reaching the maximum number of iterations.

3) Runtime Comparison: To further evaluate the efficiency of the proposed methods, we conducted a runtime comparison with baseline algorithms on the IP, Salinas, and WHU-LK. Table IV summarizes the average runtime (in seconds) of different algorithms. The results demonstrate that 3DSS-ILRQ achieves the fastest runtime across all datasets, with 32.4, 61.6, and 79.6 s for IP, Salinas, and WHU-LK, respectively. This highlights its computational efficiency. On the contrary, MTDL and DP-TRPCA exhibit significantly higher runtimes, particularly on larger datasets like WHU-LK, where MTDL takes 5526.2 s, and DP-TRPCA takes 2558.5 s. The proposed 3DSSS-ILRQ also shows competitive performance under additional superspatial units, with 90.8, 1126.6, and 477.2 s runtime for IP, Salinas, and WHU-LK, respectively. These results indicate that our methods achieve superior classification performance and maintain computational efficiency, making them suitable for practical applications.

4) Hyperparameter Sensitivity: The number of superspatial-rows s^w, the number of superspatial-columns s^q, and the number of superspectral units s^p are the primary hyperparameters of 3DSS-ILRQ and 3DSSS-ILRQ. In order to reflect their influence and conduct parameter tuning, we make a detailed comparison of their performance under different sets in Fig. 11. As shown in Fig. 11, the accuracy of 3DSS-ILRQ commonly increases as s^p increases until near the upper limit. Therefore, the value of s^p on IP, Salinas, and WHU-LK is set as 40, 55, and 50, respectively. Similarly, according to the classification results, the setting of (s^w, s^q, s^p) in 3DSSS-ILRQ can be fixed at (10,10,40), (2,4,55), and (14,14,50) on IP, Salinas, and WHU-LK, respectively.

5) Robustness to Noise Intensity and Classifier Independence: To further evaluate robustness under intensified noise and verify classifier independence, we conduct additional experiments on the IP dataset. S^2Mamba and 3DSS-ILRQ, the two top-performing methods in Table I, are assessed under elevated noise levels with both SVM and K-nearest neighbors (KNNs) with Euclidean distance. The intensified

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parameters are $\sigma = 0.10$ in Case 1, SNR of range [5,15] dB in Case 2, and 60 bands with 70%–90% impulse rates in Case 4. As shown in Table V, 3DSS-ILRQ consistently surpasses S^2Mamba under most scenarios and both classifiers. The consistent ranking between SVM and KNN confirms that the superiority of 3DSS-ILRQ is classifier-agnostic rather than an artifact of a specific classifier. These trends, aligned with those under standard noise intensities in Table I, corroborate the framework's scalable robustness.

V. Conclusion and Future Work

This article elaborates on a 3DSS-ILRQ and its extended version, 3DSSS-ILRQ, for classifying noisy HSIs. The proposed models innovatively extend the identification of perceptual superunits from the spatial domain to the spectral dimension, thereby enhancing the low-rank feature extraction capabilities of ILRQ. They effectively identify 3-D spatial and spectral superunits to generate structure-protected homogeneous subcubes while naturally eliminating redundant features through intelligent utilization of the acquired subcubes. A series of experiments on multiple simulated datasets derived from three real-world HSIs consistently confirms the proposed models' superior robustness and classification effectiveness. Looking ahead, the development of a morphologically and directionally adaptable multiscale superunit identification paradigm represents an auspicious direction for future research. In addition, we plan to investigate model optimization strategies to enhance flexibility for interdisciplinary applications across various scientific domains.

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